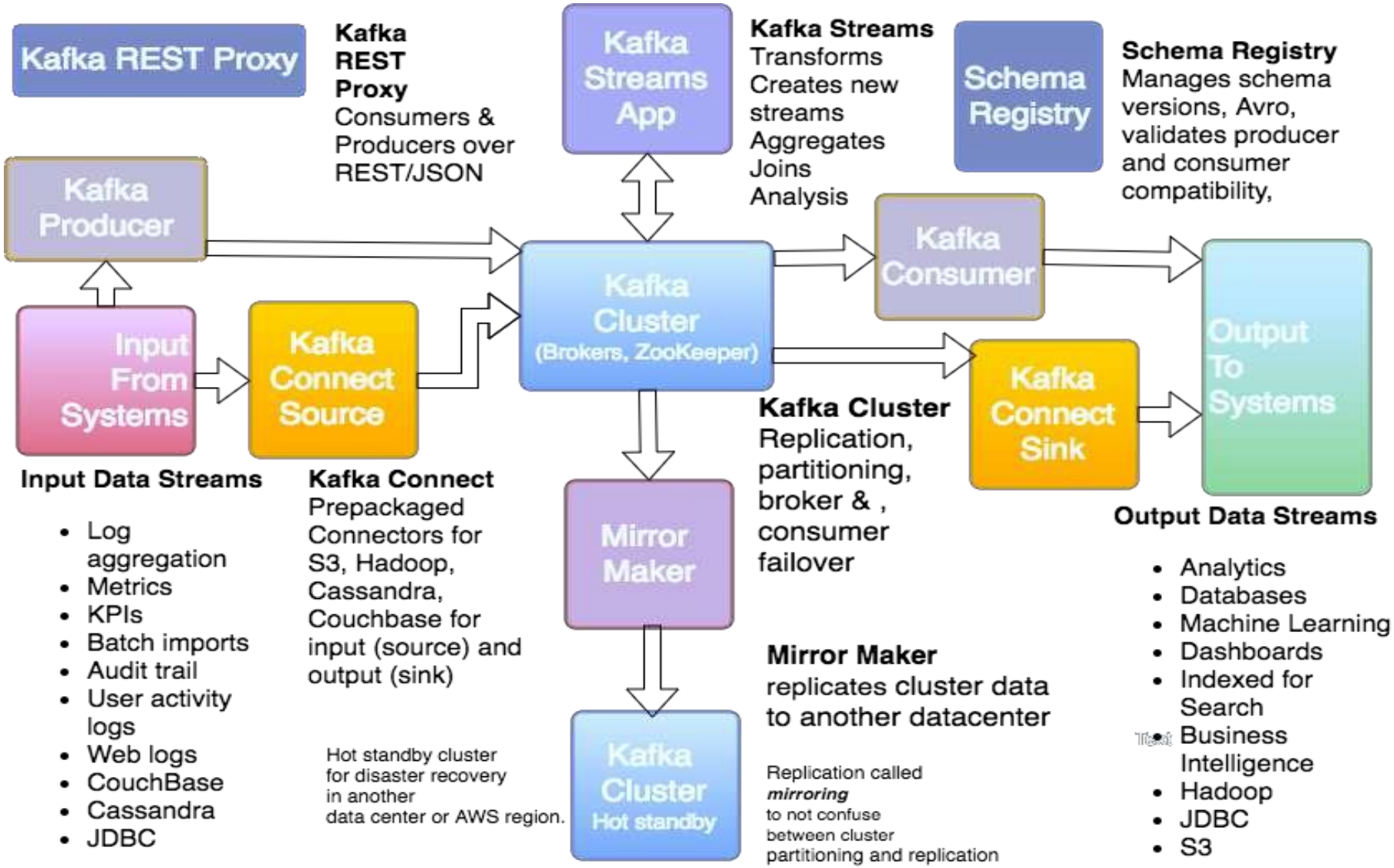
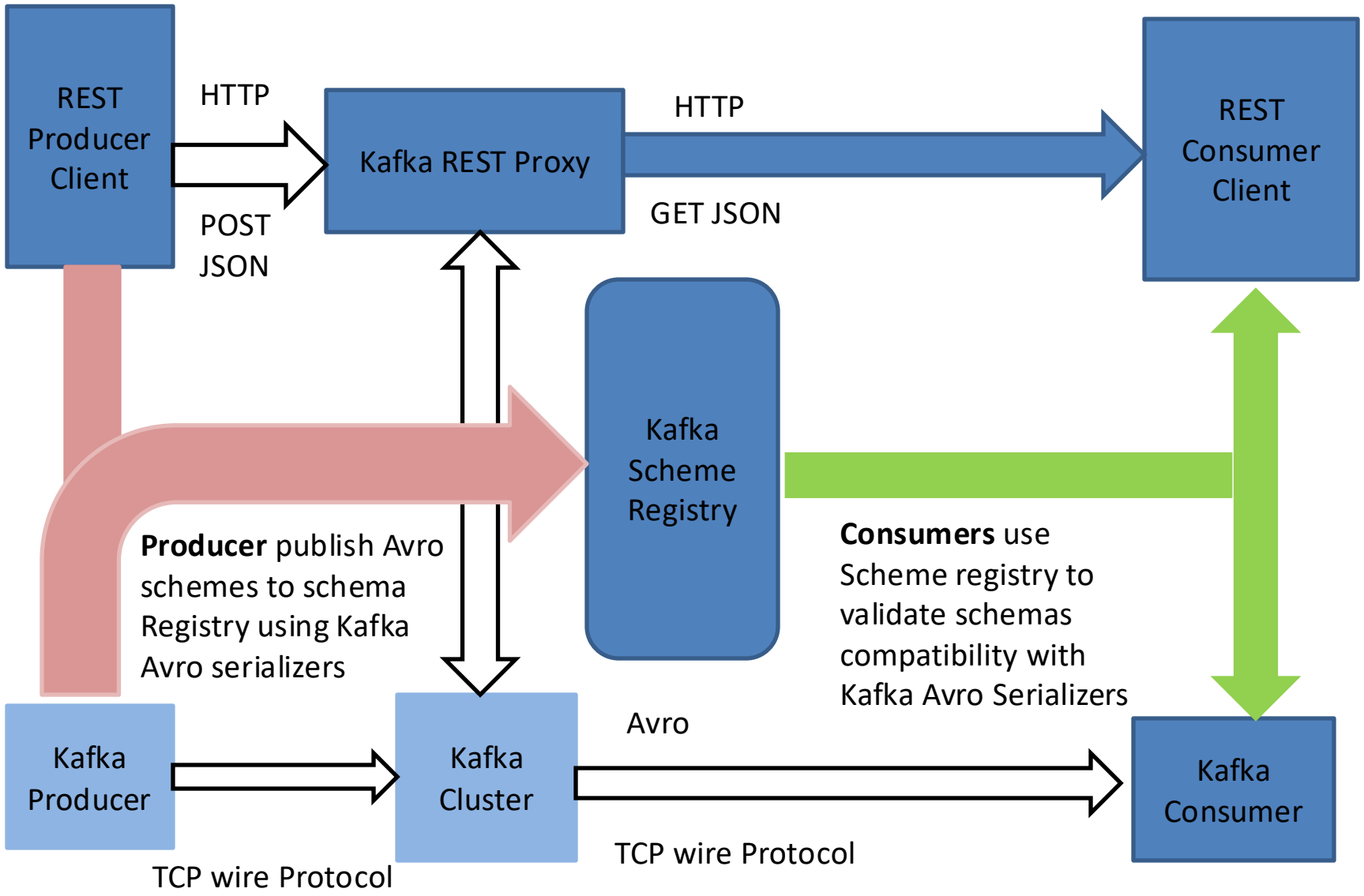
Kafka Ecosystem

Kafka Universe



Kafka REST Proxy and Schema Registry

Tos



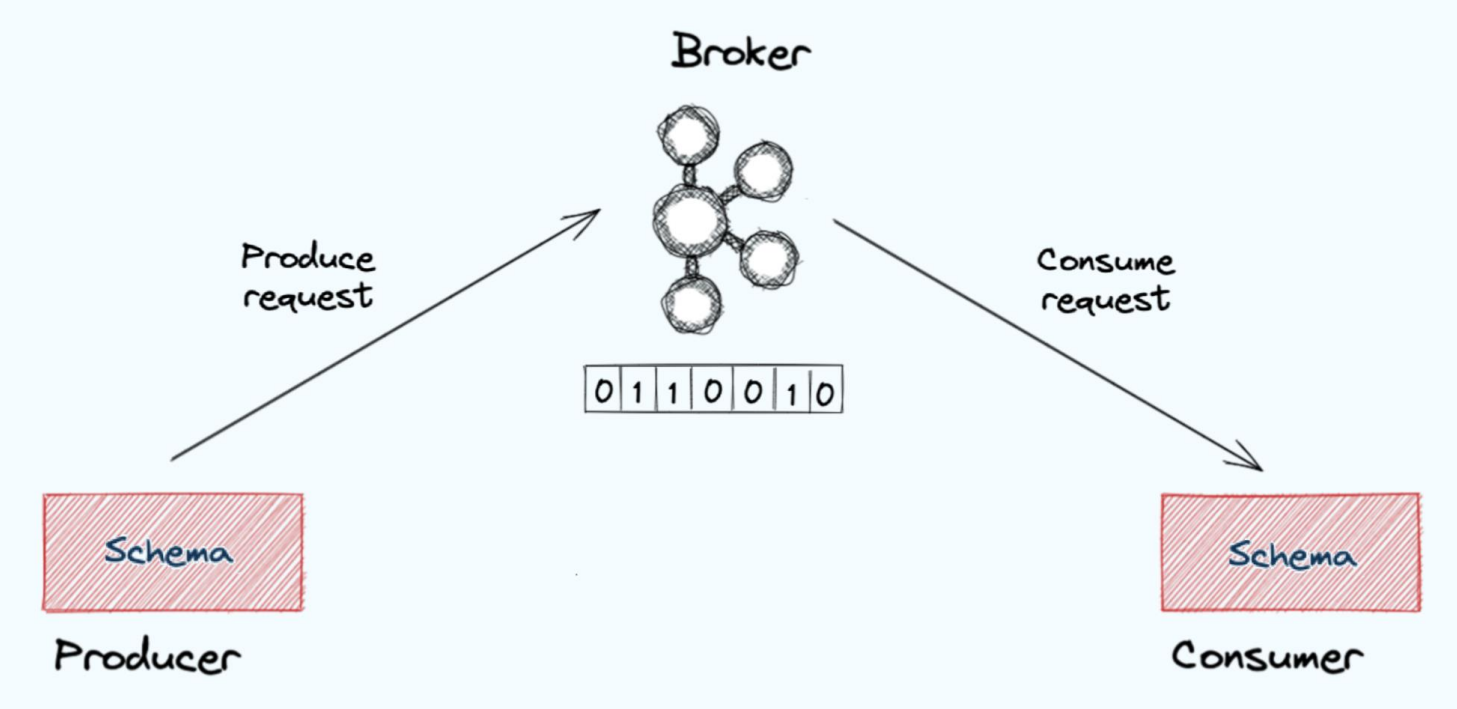
Tos

Schema Registry

Confluent Schema Registry

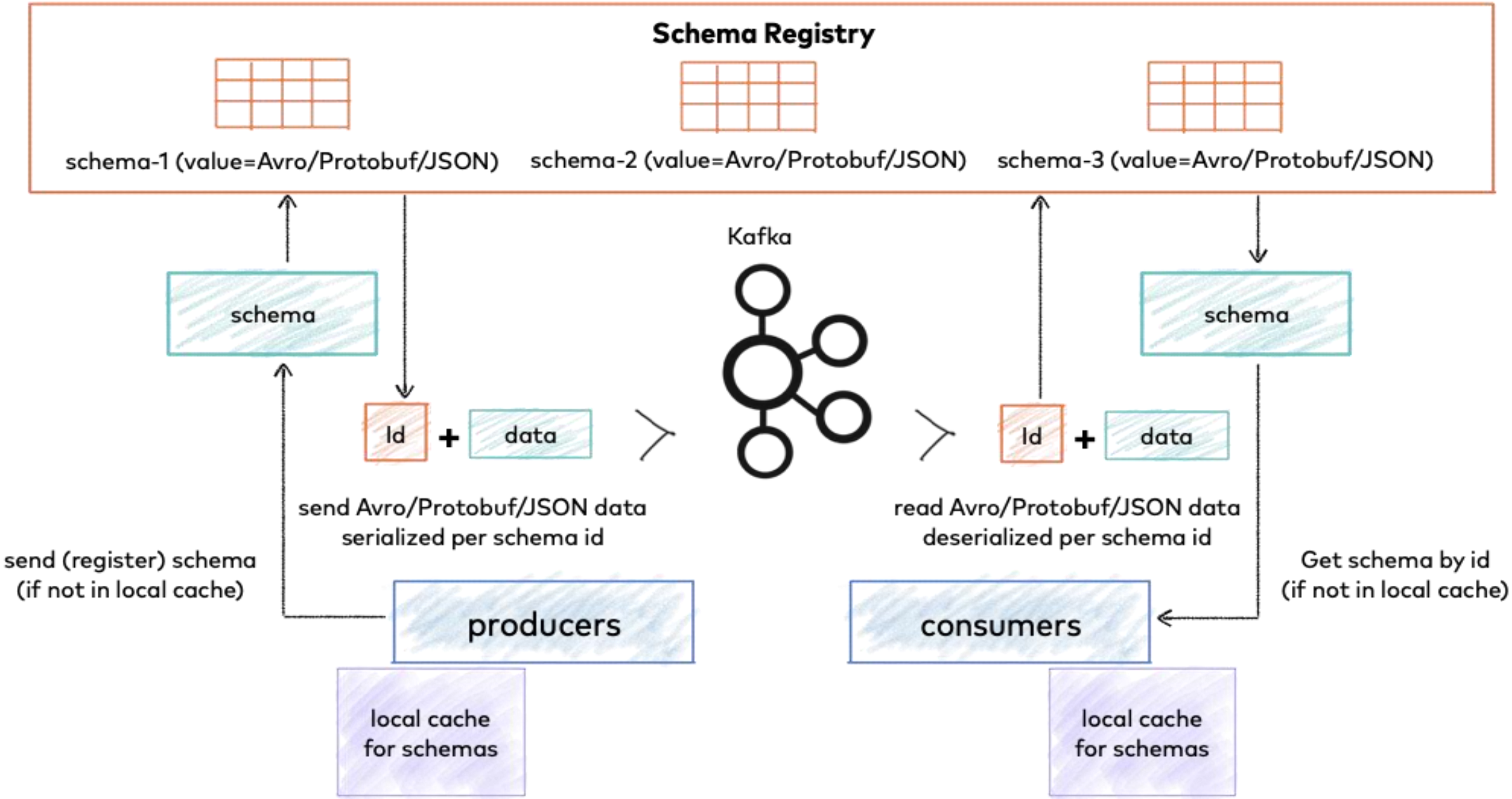
Confluent Schema Registry provides a serving layer for your metadata.
It provides a RESTful interface for storing and retrieving your Avro®, JSON Schema, and Protobufschemas.
It stores a versioned history of all schemas based on a specified subject name strategy, provides multiple compatibility settings and allows evolution of schemas according to the configured compatibility settings and expanded support for these schema types.
It provides serializers that plug into Apache Kafka® clients that handle schema storage and retrieval for Kafka messages that are sent in any of the supported formats.

Schema Is the Contract



If either your producer or consumer fails to meet the rules established by the contract, there needs to be a consequence.
This contract is codified in something called a schema.
A schema is a set of rules that establishes the format of the messages being sent.
It outlines the structure of the message, the names of any fields, what data types they contain, and any other important details.
This schema is a contract between the two applications

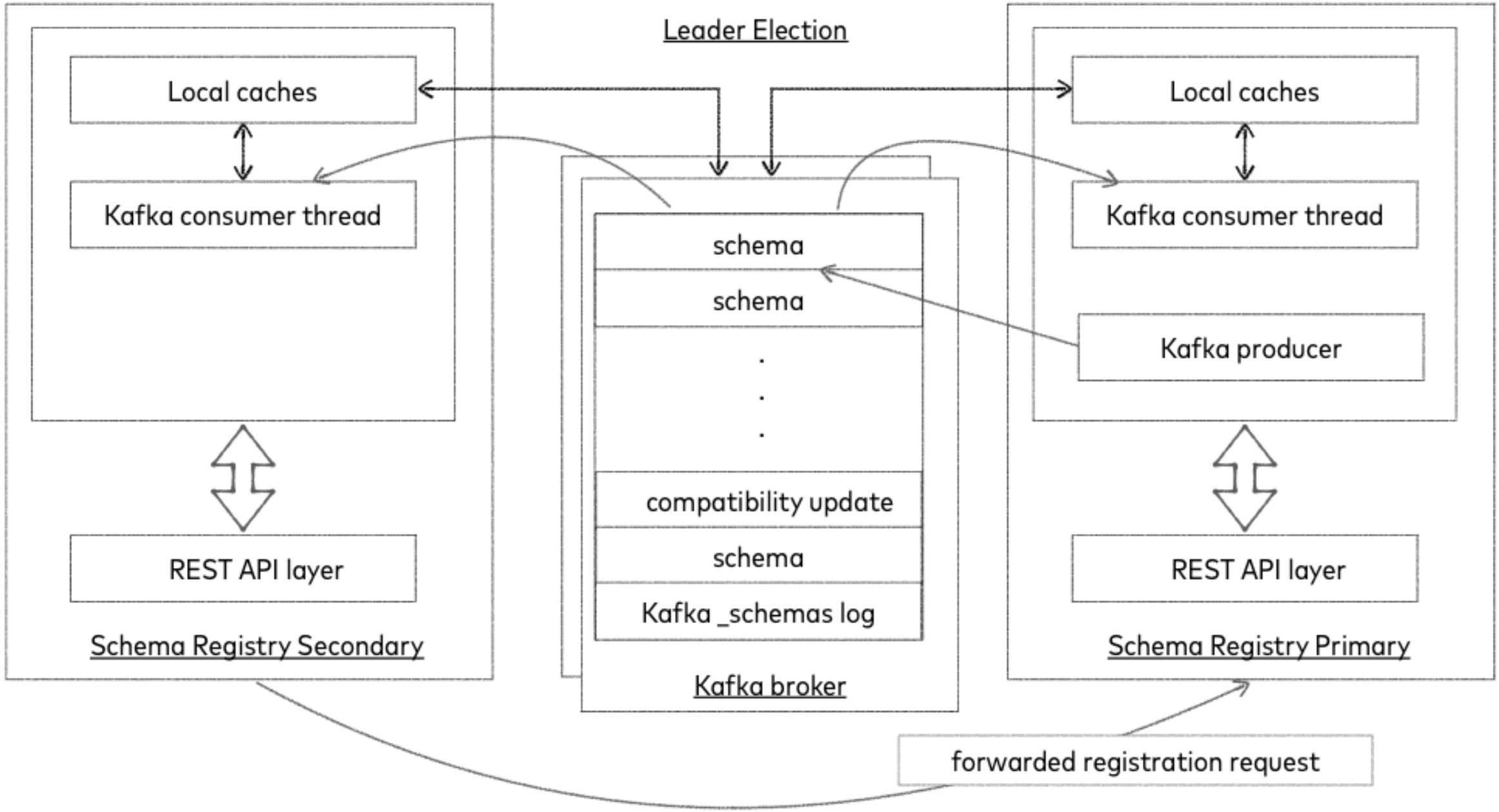
Confluent Schema Registry



Schema Registry

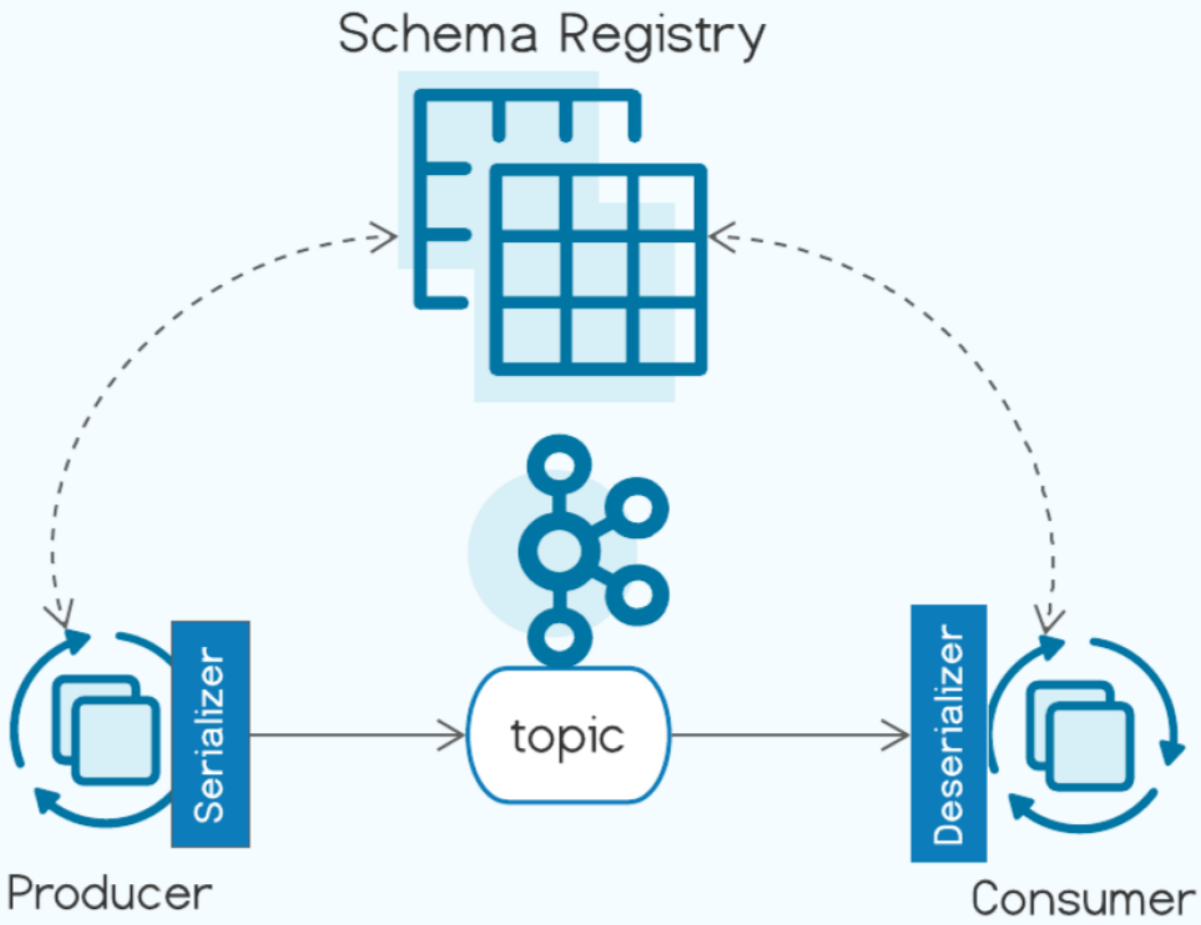
The schema registry is a service that records the various schemas and their different versions as they evolve.
Producer and consumer clients retrieve schemas from the schema registry via HTTPS, store them locally in cache, and use them to serialize and deserialize messages sent to and received from Kafka.
This schema retrieval occurs only once for a given schema and from that point on the cached copy is relied upon.

Confluent Schema Registry - **Single Primary Architecture**



Confluent Schema Registry - **Single Primary Architecture**

- **Versioned** schema repository
- **Safe** schema evolution
- **Resilient** data pipelines
- **Enhanced** data integrity
- **Reduce** storage and computation
- **Discover** your data
- **Cost-efficient** ecosystem



AVRO protobuf Protocol Buffers {JSON}

Understanding the Schema Registry Workflow

The workflow of Schema Registry includes:

- Writing a schema file
- Adding schema files to a project
- Leveraging schema tools and plugins—Maven and Gradle
- Configuring Schema Registry plugins
- Generating the model objects from a schema
- Locating and exploring the generated files

Schema file - Protobuf

```
syntax = "proto3";
package io.confluent.developer.proto;
option java_outer_classname = "PurchaseProto";

message Purchase {
  string item = 1;
  double total_cost = 2;
  string customer_id = 3;
}
```

- The first line declares the version of Protocol Buffers to use. This course uses version 3.
- The ***package*** declaration creates a name space for proto files to prevent name clashes. It ends up being the package name in the generated Java class unless you specify an ***option java_package*** in the proto file.
- The ***option java_outer_classname*** field describes the name of the generated java file
- Protocol Buffers define the ***message*** in the schema as an inner class in the generated file

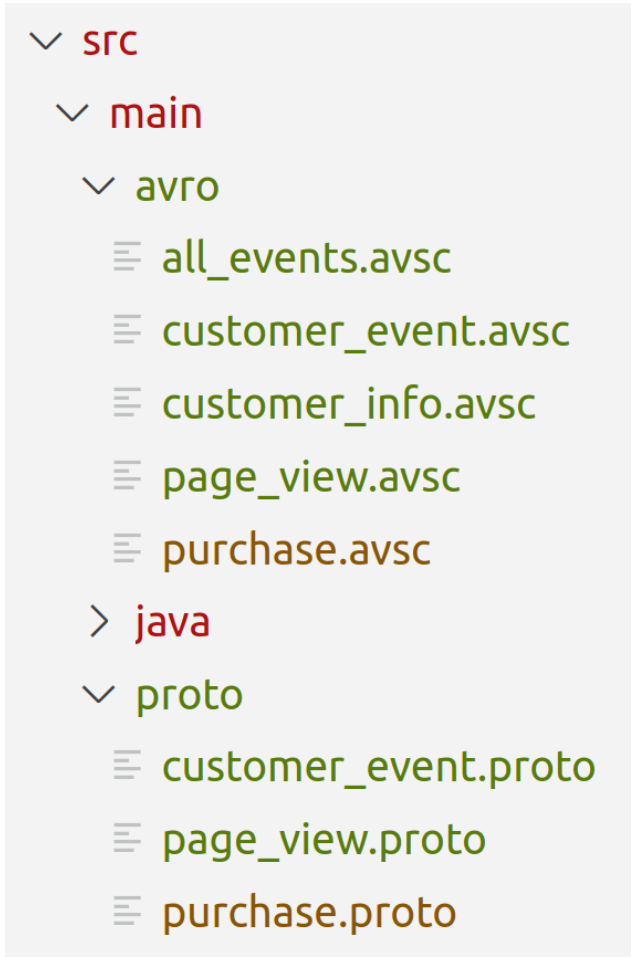
Schema file - Avro

```
{
  "type":"record",
  "namespace": "io.confluent.developer.avro",
  "name":"Purchase",
  "fields": [
    {"name": "item", "type":"string"},
    {"name": "total_cost", "type": "double"},
    {"name": "customer_id", "type": "string"}
  ]
}
```

- The ***namespace*** field serves the same purpose as in the Protobuf file, preventing name collisions. The namespace also becomes the package name for the generated Java file.
- Fields are declared in a ***JSON*** array.

Schema Files in a Project

- The Avro schema files go in the ***src/main/avro*** directory and ***Protobuf*** files land in ***src/main/proto***.

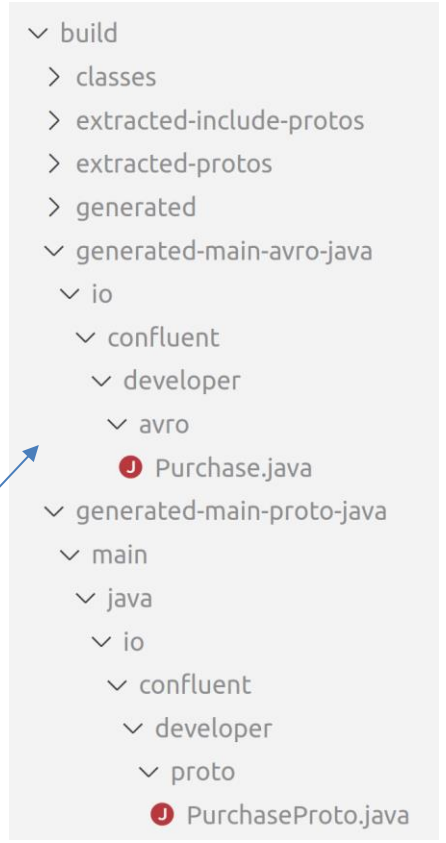


Once you've written your schema files you will want to generate the model/data objects from the schema files. Fortunately, there are plugins available, either Maven or Gradle, that can integrate into your local build.

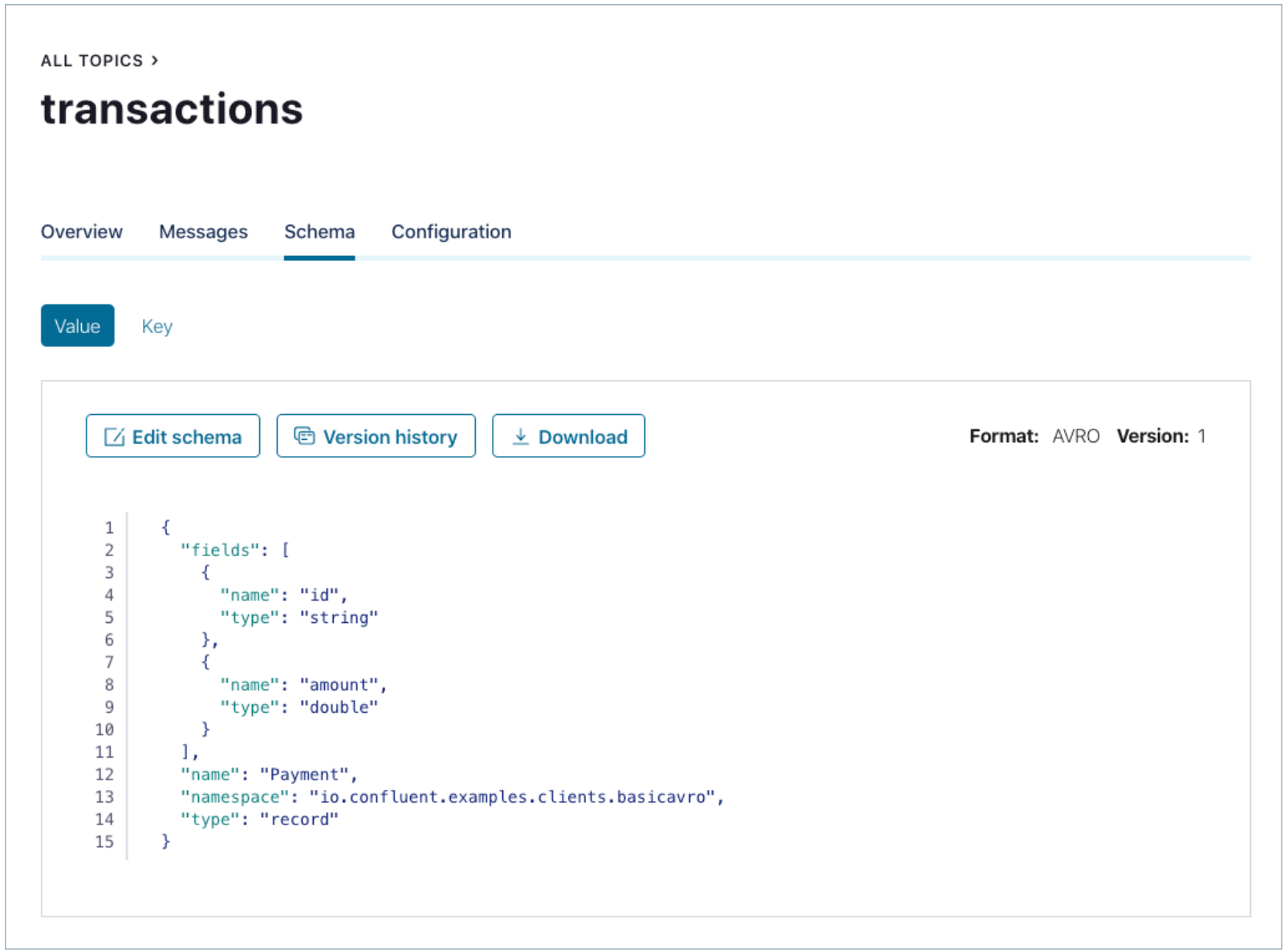
```
schemaRegistry {
    url = Confluent Cloud SR endpoint
    credentials {
        //characters up to the ':' in the basic.auth.user.info property
        username = <username>
        // password is everything after ':' in the basic.auth.user.info property
        password = <password>
    }
    // Possible types are ["JSON", "PROTOBUF", "AVRO"]
    register {
        subject('avro-purchase', 'src/main/avro/purchase.avsc', 'AVRO')
        subject('proto-purchase', 'src/main/proto/purchase.proto', 'PROTOBUF')
    }
}
```

Plugins to generate files

Generated Class files



Confluent Schema Registry - **Single Primary Architecture**



Take note of the version, schema ID, and the fields.

Confluent Schema Registry - **Single Primary Architecture**

etc/schema-registry/schema-registry.properties

```
# Specify the address the socket server listens on, e.g. listeners = PLAINTEXT://your.host.name:9092
listeners=http://0.0.0.0:8081

# The host name advertised in ZooKeeper. This must be specified if your running Schema Registry
# with multiple nodes.
host.name=192.168.50.1

# List of Kafka brokers to connect to, e.g. PLAINTEXT://hostname:9092,SSL://hostname2:9092
kafkastore.bootstrap.servers=PLAINTEXT://hostname:9092,SSL://hostname2:9092
```

Configuring Schema Registry API

Kafka applications using Avro data and Schema Registry need to specify at least two configuration parameters:

- Avro serializer or deserializer
- Properties to connect to Schema Registry

```
...
import io.confluent.kafka.serializers.KafkaAvroSerializer;
...
props.put(ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG, StringSerializer.class);
props.put(ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG, KafkaAvroSerializer.class);
...
KafkaProducer<String, Payment> producer = new KafkaProducer<String, Payment>(props));
final Payment payment = new Payment(orderId, 1000.00d);
final ProducerRecord<String, Payment> record = new ProducerRecord<String, Payment>(TOPIC, payment
.getId().toString(), payment);
producer.send(record);
...
```

```
...
import io.confluent.kafka.serializers.KafkaAvroDeserializer;
...
props.put(ConsumerConfig.KEY_DESERIALIZER_CLASS_CONFIG, StringDeserializer.class);
props.put(ConsumerConfig.VALUE_DESERIALIZER_CLASS_CONFIG, KafkaAvroDeserializer.class);
props.put(KafkaAvroDeserializerConfig.SPECIFIC_AVRO_READER_CONFIG, true);
...
KafkaConsumer<String, Payment> consumer = new KafkaConsumer<>(props));
consumer.subscribe(Collections.singletonList(TOPIC));
while (true) {
  ConsumerRecords<String, Payment> records = consumer.poll(100);
  for (ConsumerRecord<String, Payment> record : records) {
    String key = record.key();
    Payment value = record.value();
  }
}
...
```

Schema Management

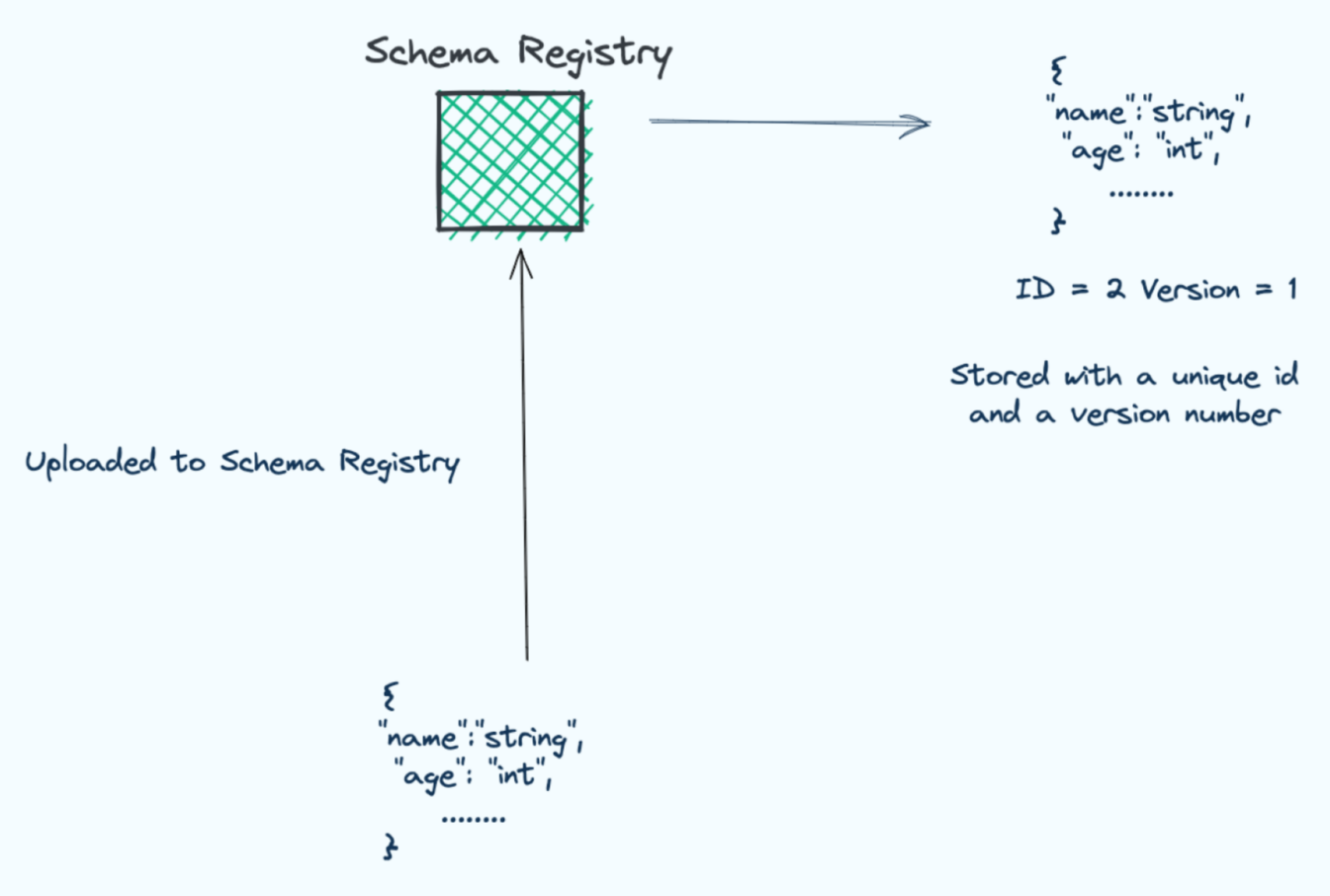
Tos

Effective schema management requires:

- Schema IDs
- Schema registration
- Schema versioning
- Viewing and retrieving schemas

Schema ID and Version

Tos



Schema management starts with registering schemas.
It also includes updating schemas and viewing or downloading them.

There are multiple ways to register a schema :
Confluent CLI, Schema Registry REST API, Confluent Cloud Console, or the Maven and Gradle plugins

Register an Avro Schema - REST API

Tos

```
jq  '. | {schema: tojson}' purchase.avsc |  curl -X POST -H \
 "Content-Type: application/vnd.schemaregistry.v1+json" \
 http://localhost:8081/subjects/my-kafka-value/versions    -d
@-
```

purchase.avsc

```
{
 "type":"record",
 "namespace": “tos.developer.avro",
 "name":"Purchase",
 "fields": [
  {"name": "item", "type":"string"},
  {"name": "total_cost", "type": "double" },
  {"name": "customer_id", "type": "string"}
 ]
}
```

```
[root@kafka0 scripts]# jq  '. | {schema: tojson}' purchase.avsc |  curl -X POST -H \
>  "Content-Type: application/vnd.schemaregistry.v1+json" \
>  http://localhost:8081/subjects/my-kafka-value/versions    -d @-
{"id":2}[root@kafka0 scripts]#
```

#dnf install jq

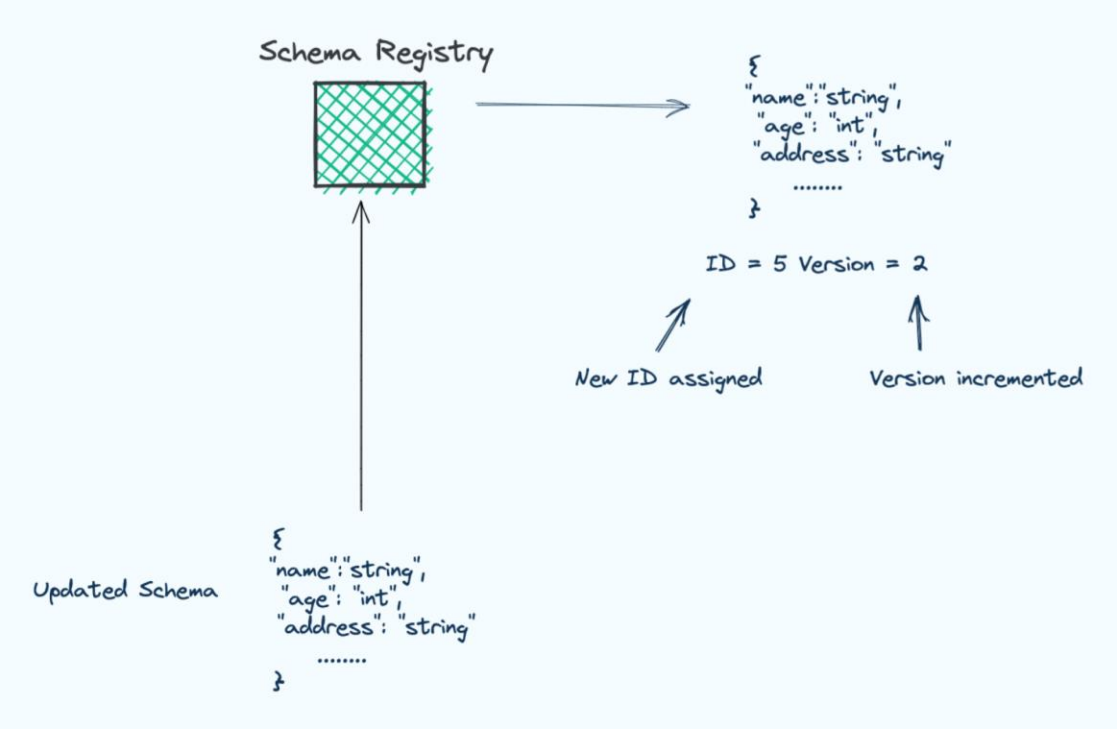
Schema Registration

Tos

When you register a schema you need to provide the ***subject-name*** and the ***schema*** itself.
The s***ubject-name*** is the ***name-space*** for the schema, almost like a key when you use a hash-map.
The standard naming convention is ***topic-name-key*** or ***topic-name-value***.
Once Schema Registry receives the ***schema***, it assigns it a unique ID number and a version number.
The first time you register a schema for a given ***subject-name***, it is assigned a version of 1.

Updating a Schema

Tos



To update previously registered schemas.
Use the same methods that was used to register the schema.

Provided you are making compatible changes (compatibility is covered in an upcoming module), Schema Registry will simply assign a new ID to the schema and a new version number.

The new ID is not guaranteed to be in sequential order.

curl -s "http://localhost:8081/schemas/ids/2" | jq

But the version number will always be incremented by one, hence it will be in sequential order

```
[root@kafka0 scripts]# curl -s "http://localhost:8081/schemas/ids/2" | jq
{
  "schema": "{\"type\":\"record\",\"name\":\"Purchase\",\"namespace\":\"tos.developer.avro\",\"fields\":[{\"name\":\"item\",\"type\":\"string\"},{\"name\":\"total_c
ost\",\"type\":\"double\"},{\"name\":\"customer_id\",\"type\":\"string\"}]}"
}
[root@kafka0 scripts]#
```

Understanding Schema Subjects

Tos

- When you register a schema you use a subject name to create a namespace or handle for the schema in Schema Registry.

  The schema subject name is used for:

  Compatibility checks—they are done per subject
  Linking version numbers to a subject

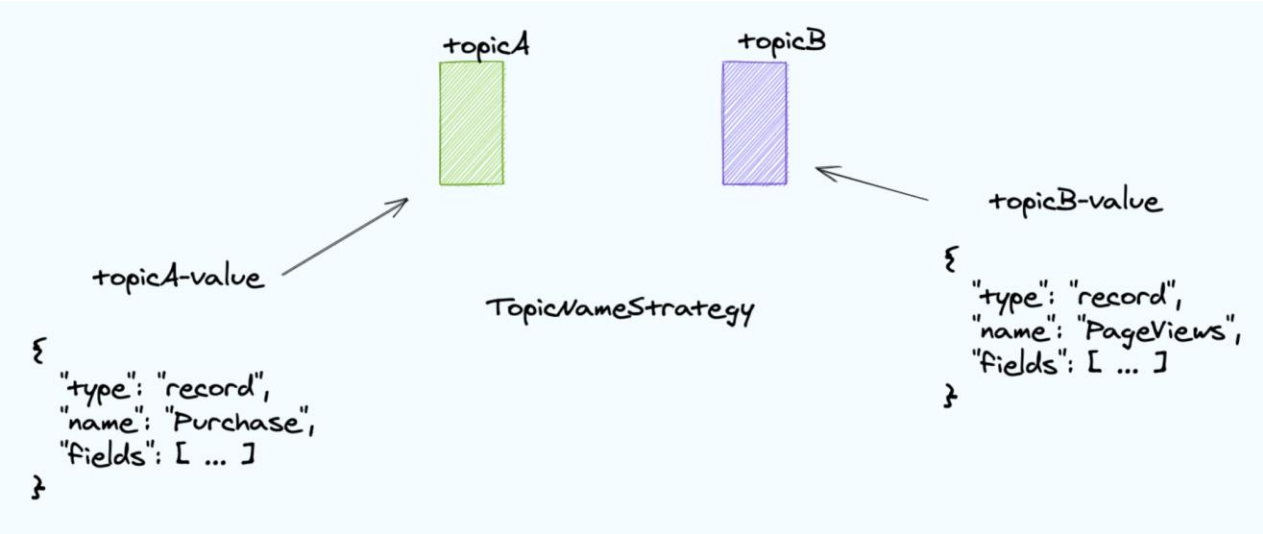
  Also when you evolve a schema by making changes to it, it's assigned a new ID and version number, but the subject stays the same.

  Essentially, the subject name is a unique identifier for a schema – a reference to retrieve it from the store.

Subject Name Strategies

Tos

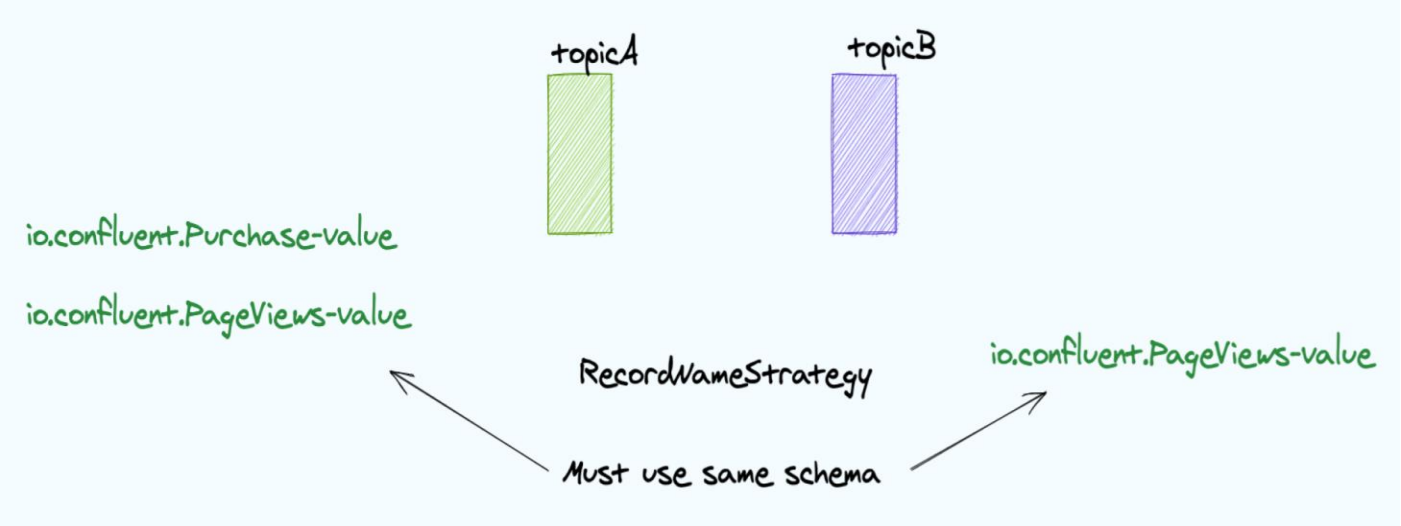
- When providing a subject name, there are three strategies or ways to provide it:
  - TopicNameStrategy (default setting)
  - RecordNameStrategy
  - TopicRecordNameStrategy



Using TopicNameStrategy effectively limits the topic to one record type, since all records in the topic must adhere to the same schema. Trying to register a schema for a different type would break the compatibility checks.

RecordNameStrategy

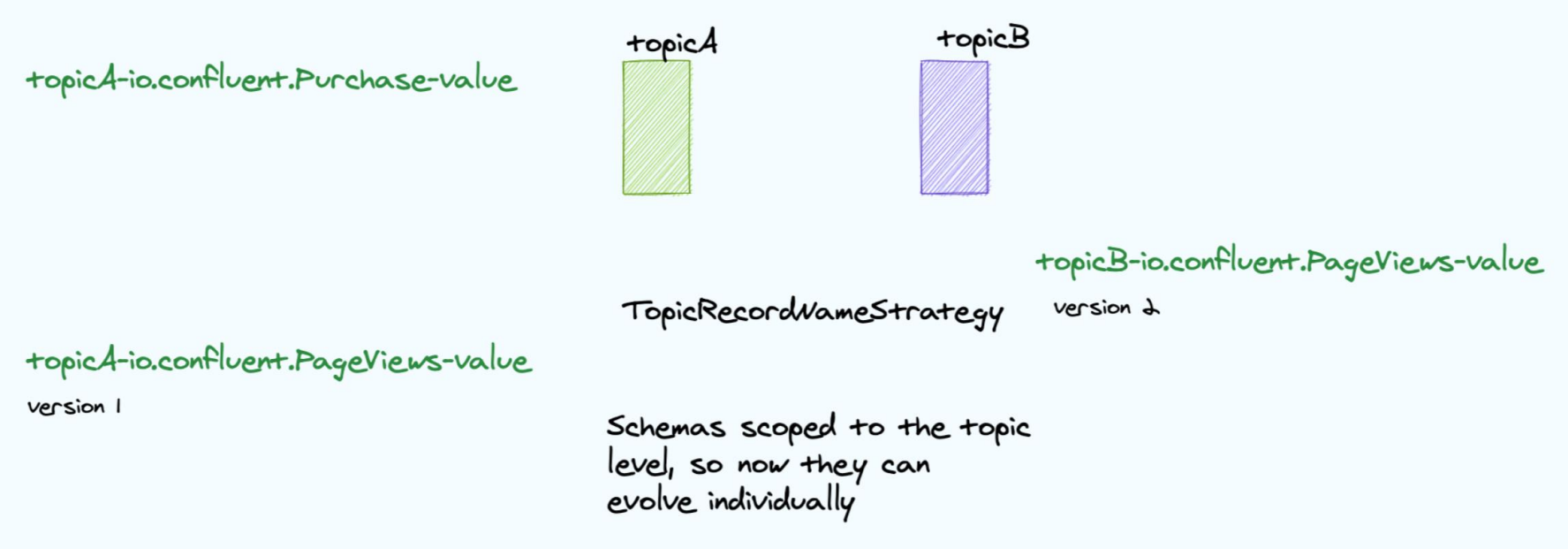
Tos



To get around the limitation of one record type for a given topic, we can use ***RecordNameStrategy***.
It creates the subject name using the fully qualified class name of the record – again followed by key or value depending on whether the schema applies to the key or value of the message.
RecordNameStrategy allows for different schemas in a topic since the individual records only need to comply with a schema that has the subject name that matches its class.
But you have to use the same schema and version across all the topics in a cluster FOR THAT PARTICULAR RECORD TYPE, since there's no way to tell which topic the record belongs to.

TopicRecordNameStrategy

Tos



TopicRecordNameStrategy is a combination of the first two strategies.
Subject names are created using ***<topic name>-<fully qualified record name>*** appended with ***-key or -value***.
Since you've now scoped the schema to a particular topic you don't have to use the same version across all topics in the cluster. FOR THAT RECORD TYPE you can evolve the schemas separately.

Using Different Subject Name Strategies

Tos

- Choosing and using a particular strategy involves two steps:
  - Registering the schema with the correct subject name format.
  - Setting the subject strategy on the clients.

Since **TopicNameStrategy** is the default, clients are already set to use it. To use a strategy other than the default,
set **key.subject.name.strategy** or **value.subject.name.strategy** on the client as needed.

Record Name Strategy -
io.confluent.kafka.serializers.subject.RecordNameStrategy
Topic Record Name Strategy -
io.confluent.kafka.serializers.subject.TopicRecordNameStrategy

Lab : Schema Registry - Manage Schemas for Topics